جامعة عبدالله السالم
Abdullah Al Salem University

Presented by : Retaj Bader , Nada Saeed

ID : 2231133203 , 2231171802

Department : Robotics & Mechatronics Engineering

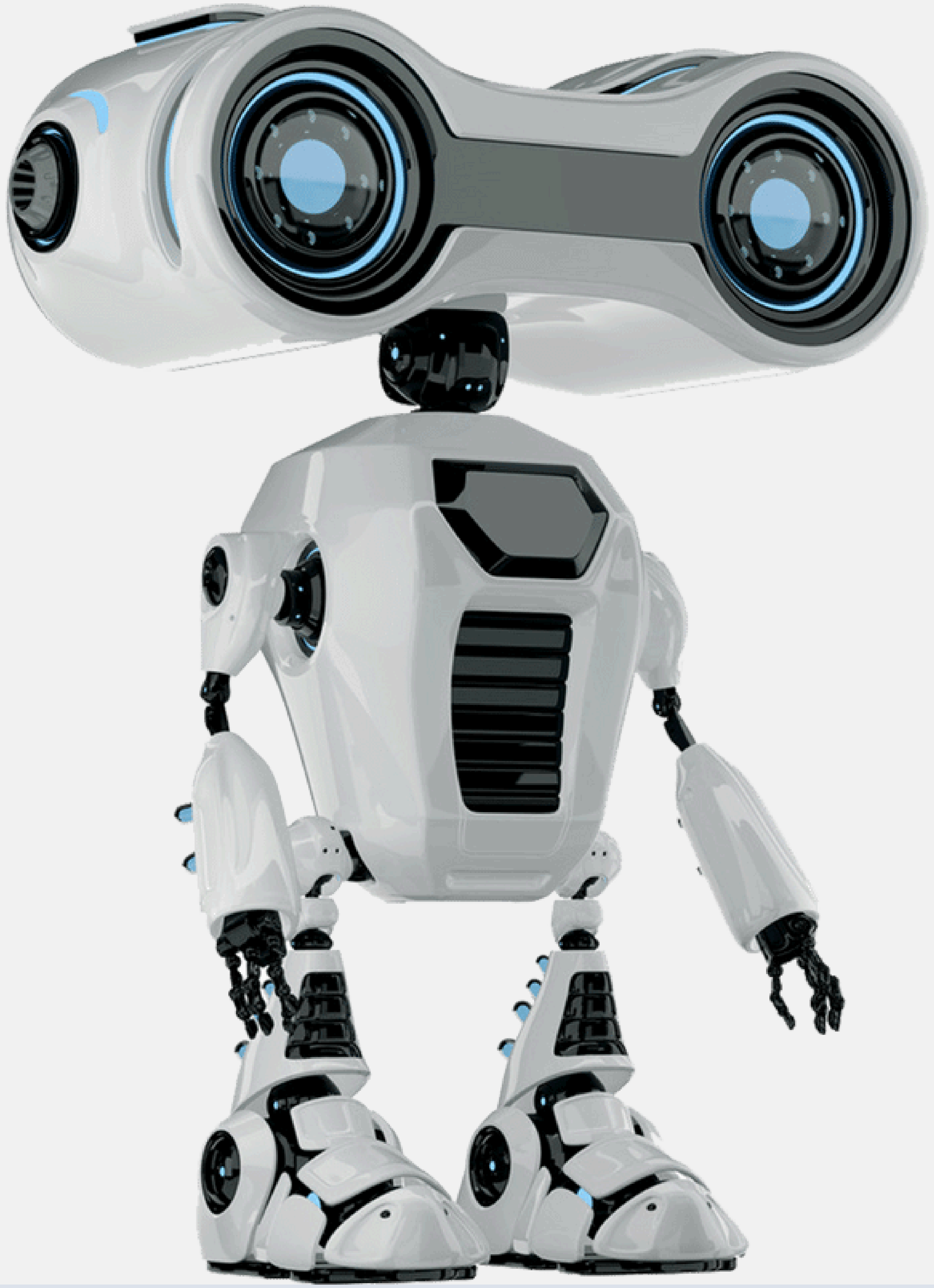
RME301 : Introduction to Mechatronics & Robotics

Spring 2025/2026

Instructor : Dr. Jamshed Iqbal

Smart Path:

Robot for color detection, obstacle avoidance and line following



Contents (Agenda)

Objectives

To develop an autonomous robot using sensors and control logic.

To implement line following, obstacle avoidance, and color detection.

To apply programming skills using Python on the mBlock platform

How the robot works?

- Codey Rocky is a pre-built robot
- It include sensor
- It is programmed using mBlock
- Python was used for implementing control logic

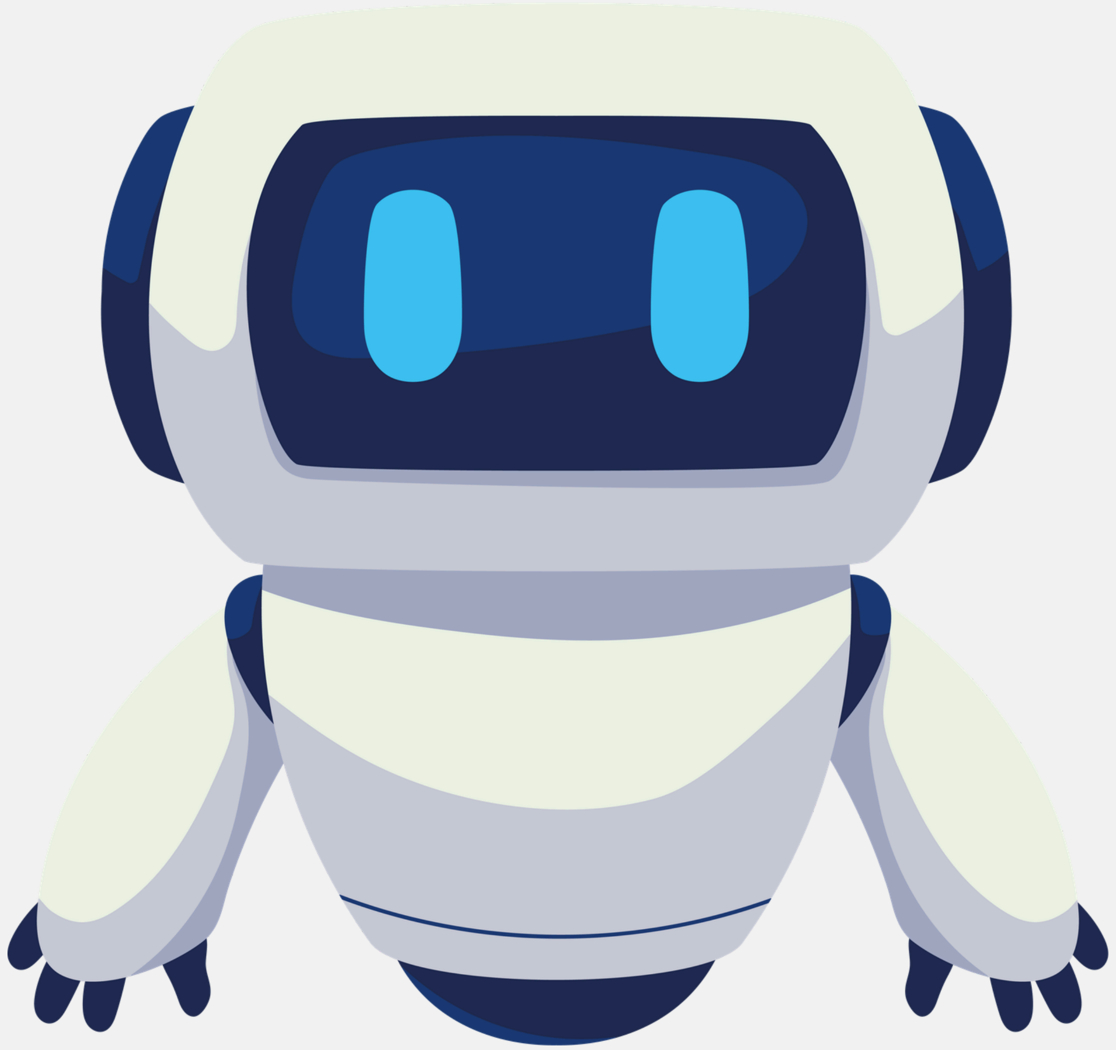
- The robot uses sensors to detect its environment
- It processes inputs using programmed logic in Python
- Also , three different functions should be done by the robot
- The robot works

Button A – Color Detection

Button B- Avoiding obstacles

Button C - Line Following

A - Color dedicate



How It works?

- Activated by pressing button A
- The robot uses the color sensor to detect objects
- Once the sensor reads a color, the robot displays both the color and its name on the screen
- The robot can detect the following colors: red , green , yellow, blue, cyan, purple, black , white .

B- Obstacle Avoidance

How it works?

- Activated by pressing button B
- The robot checks for obstacles using the distance sensor
- If an obstacle is detected:
    - Stop
    - Turn left (90°)
    - Move forward
- If no obstacle:
    - Move forward

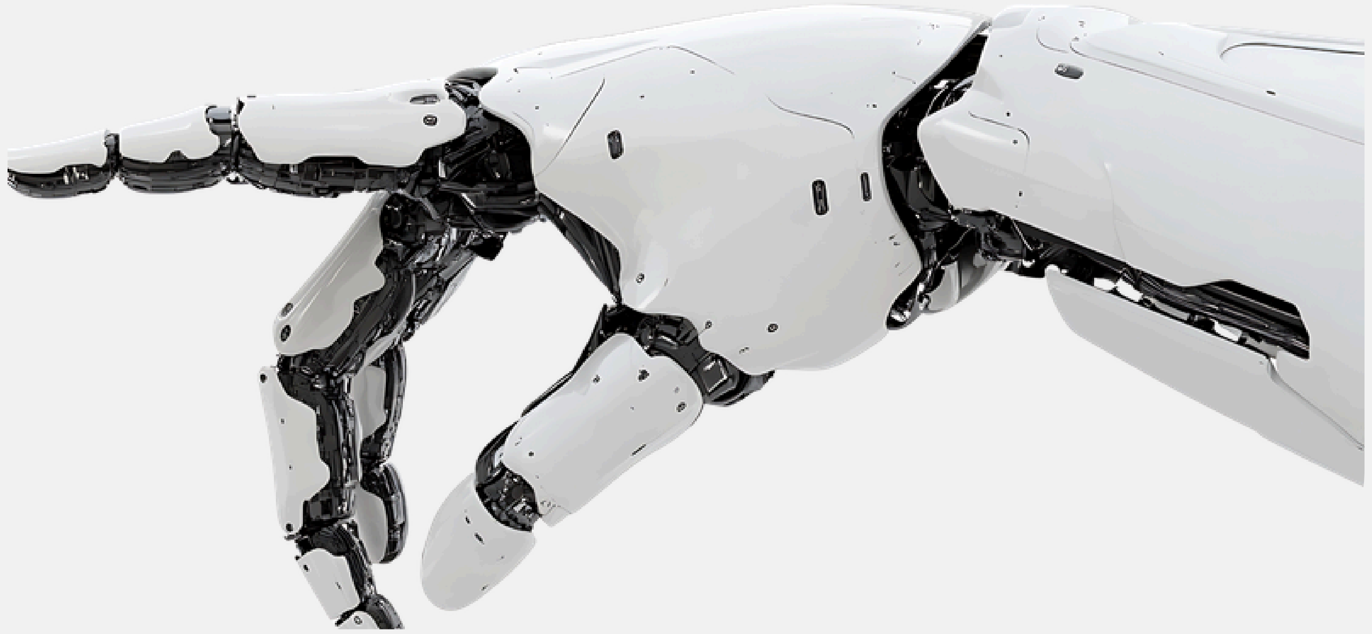
C-Line Following

How it works?

- Activated by pressing button C
- The robot reads the line sensor
- Based on line position:
  - Center → Move forward
  - Left → Turn left
  - Right → Turn right

Results

- The robot successfully performed all tasks
- Line following worked with good accuracy
- Obstacle avoidance prevented collisions
- Color detection correctly identified colors



Challenges

- Sensor sensitivity issues

- Difficulty in handling multiple functions separately
- Lighting affected color detection
- Limited experience with Python
- The curved line path required additional time for tuning and adjustment

Skills Gained & Improvements

Skills Gained:

- Improved Python programming skills
- Better understanding of sensors
- Problem-solving and debugging skills

Improvements:

- Better time management
- Improved sensor calibration
- More structured coding

References

Makeblock Co., Ltd. (n.d.). *Codey Rocky user guide*. https://www.makeblock.com

Makeblock Co., Ltd. (n.d.). *mBlock software documentation*. https://www.mblock.cc

Miro. (n.d.). *Miro: Online collaborative whiteboard platform*. https://miro.com/

Siegwart, R., Nourbakhsh, I. R., & Scaramuzza, D. (2011). *Introduction to autonomous mobile robots* (2nd ed.). MIT Press.

YouTube. (n.d.). *Line following robot using Codey Rocky* [Video]. YouTube. https://youtu.be/zNoaZdvFefE

YouTube. (n.d.). *Obstacle avoidance using Codey Rocky* [Video]. YouTube. https://youtu.be/7xJflVm1NPQ